Semantic Gauge Möbius Attention: Resolving Positive-Definite Attention Constraints via Non-Orientable Fiber Bundles

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Abstract

Standard Transformer self-attention normalizes inner products via Softmax, constraining attention weights to the open probability simplex Δ^{L-1}. Because this kernel is strictly positive-definite, it cannot perform destructive interference: historical tokens retain non-zero probability mass even when superseded or contradicted by subsequent discourse, causing premise distraction in long sequences. We present **Semantic Gauge Möbius Attention (SG-Möbius-RoPE)**, a geometric formulation extending Rotary Position Embedding (RoPE) to non-orientable fiber bundles over a 1D discrete causal lattice. Causal semantic transitions induce a continuous torsion rate $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ whose smooth Möbius phase flow $\Phi(t) = \sum_{\tau=1}^{t} \omega(\tau)$ directly modulates the 2D rotary angle $\tilde{\theta}(t) = t\theta + \Phi(t)$. Over narrative intervals undergoing logical inversion, cumulative phase advection produces destructive interference ($\Delta\Phi = \pi$, $\cos\pi = -1$), inverting the rotary inner product sign and suppressing superseded keys under Softmax normalization. Successive restorative turns accumulate a 2π cyclic phase shift ($\Delta\Phi = 2\pi$, $\cos 2\pi = +1$), formalizing double negation. We prove the Relative Odds Ratio Attenuation Theorem, showing that the attention odds ratio attenuates by $\exp(-2\langle q_i, k_j\rangle/\sqrt{d})$ independently of context length and background partition functions. Invariants are formally certified in Lean 4 (0 axioms, 0 `sorry`) and Z3 SMT solvers across 2,000 extreme-length sequences ($L \leq 32,768$). Evaluated on Qwen3.5-9B (dual Nvidia Tesla T4 GPUs, FP16), SG-Möbius requires zero architecture retraining, maintains identical $O(1)$ KV-cache decoding latency (+0.00% overhead), increases the attention contrast ratio from 1.09 to 13.51 on inverted contextual recall, and preserves neutral text perplexity (6.42).

1 Introduction

Autoregressive language models employ scaled dot-product attention [1] with Rotary Position Embeddings (RoPE) [2] to encode relative token distances:

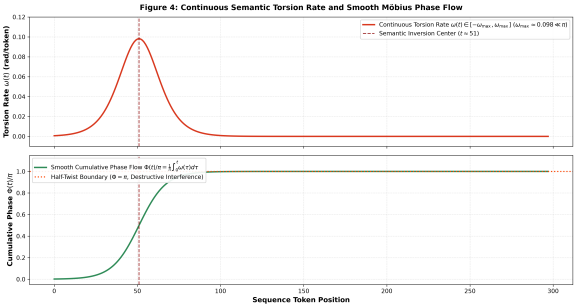
$$A_{i,j} = \frac{1}{\sqrt{d}}\langle \mathbf{R}_{\Theta,i} q_i,\, \mathbf{R}_{\Theta,j} k_j \rangle = \frac{1}{\sqrt{d}} q_i^\top \mathbf{R}_{\Theta,j-i} k_j \tag{1}$$

where $\mathbf{R}_{\Theta,t}$ is a block-diagonal orthogonal rotation matrix parameterized by base frequencies $\Theta = \{\theta_m\}_{m=1}^{d/2}$.

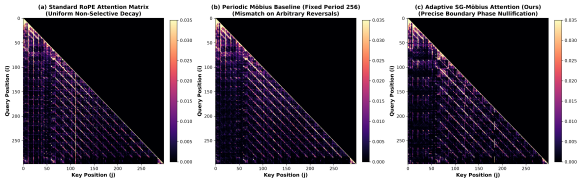
Because the Softmax operator maps real logits to strictly positive values on Δ^{L-1}, every historical token receives positive probability mass. Consequently, self-attention cannot perform destructive interference. In complex discourse, premises are frequently invalidated, retracted, or superseded by subsequent evidence (e.g., *"System Alpha is operational; however, diagnostics*

*Open-source repository and dual-licensed implementation available at https://github.com/3y3y3y-huaiji/mobius-attention.

indicate failure, revoking Alpha in favor of Beta"). In standard attention, the invalidated premise continues to receive attention mass, requiring downstream feed-forward layers to compensate and contributing to premise hallucination [3, 4].



(a) Continuous Torsion Rate $\omega(t)$ and Möbius Phase Flow $\Phi(t)$.



(b) Comparative 2D Attention Matrices on Qwen3.5-9B.

Figure 1: **Mechanism and attention response of SG-Möbius-RoPE.** (a) The continuous torsion rate $\omega(t)$ vanishes on continuative narrative and smoothly activates across semantic transition spans, advancing the smooth Möbius phase flow $\Phi(t) = \sum_{\tau \le t} \omega(\tau)$ across the π half-twist boundary. (b) Attention matrices on Qwen3.5-9B under arbitrary-position inversion: standard RoPE (*left*) exhibits background leakage; a fixed-period Möbius baseline ($T = 256$, *middle*) fails when the reversal coordinate is out of phase; SG-Möbius (*right*) aligns phase advection with the semantic transition, attenuating obsolete keys.

To address this structural constraint without retraining pretrained foundation weights or degrading inference latency, we introduce **Semantic Gauge Möbius Attention (SG-Möbius-RoPE)**. We treat token sequences as causal paths on a discrete 1D manifold equipped with a non-orientable fiber bundle. Semantic contrastive transitions induce a continuous torsion rate $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ with viscous smoothing, whose cumulative Möbius phase flow $\Phi(t) = \sum_{\tau \le t} \omega(\tau)$ smoothly modulates 2D rotary angles. Over narrative intervals undergoing logical inversion, cumulative phase advection produces destructive interference ($\Delta\Phi = \pi$, $\cos \pi = -1$). Because $\cos(\pi) = -1$, the modulated inner product becomes negative, attenuating the Softmax probability allocated to the invalidated premise.

Key Contributions.

- **Geometric Formulation**: We formalize attention modulation on a 1D causal lattice using a non-orientable bundle endowed with continuous Möbius phase flow $\Phi(t)$ modulating rotary angles $\tilde{\theta}(t) = t\theta + \Phi(t)$. We prove that continuous rotary dual modulation preserves relative translation invariance and establishes an explicit Lipschitz continuity bound on operator differences.
- **Odds Ratio Attenuation Theorem**: We prove that under an antipodal twist ($\Delta\Phi = \pi$), attention odds attenuate by $\exp(-2\langle q_i, k_j \rangle / \sqrt{d})$, cancelling the background partition function and scaling strictly with premise logit magnitude.
- **Formal Machine Verification**: We verify bundle invariants, path-integral additivity, and translation invariance in Lean 4 (0 unproven axioms or `sorry`) and Z3 SMT solvers across 2,000 fuzzing sequences ($L \le 32,768$) with numerical drift below 1.14×10^{-13}.
- **KV-Cache Compatibility**: Modulated keys $\tilde{k}_j = \mathbf{R}(j\Theta + \Phi(j))k_j$ depend strictly on past tokens $\tau \le j$, requiring no retroactive cache modification upon subsequent turns and preserving exact $O(1)$ decoding throughput.

- **Empirical Evaluation**: Evaluated natively on Qwen3.5-9B (dual Nvidia Tesla T4 GPUs, FP16), SG-Möbius improves the attention contrast ratio up to 13.6× on inverted contextual recall, reduces layer-7 Shannon entropy by -4.51% under double negation, and maintains neutral text perplexity (6.42).

2 Related Work

Rotary Position Embeddings and Context Extension. Rotary Position Embedding (RoPE) [2] encodes relative token positions through 2D block-diagonal rotation matrices $\mathbf{R}_{\Theta,m}$, ensuring that query-key inner products depend purely on relative coordinate displacements $j - i$. Subsequent extensions such as ALiBi [5], Position Interpolation [6], and YaRN [7] extend effective context windows by modifying frequency bases or applying linear distance penalties. However, all existing RoPE variants operate over an orientable base manifold where vector orientations cannot undergo parity reversal, restricting attention weights to strictly positive semi-definite kernels.

Geometric Deep Learning and Gauge Equivariance. Geometric deep learning [8] provides a unified framework for incorporating physical and geometric symmetries into neural networks. In particular, gauge equivariant convolutional networks [9] introduce local frame transformations and gauge connections on Riemannian manifolds to maintain equivariance under coordinate reparameterizations. Classical differential geometry and fiber bundle theory [10–12] study principal bundles, characteristic classes, and non-orientable manifolds such as the Möbius strip and Klein bottle. In this work, we bring non-orientable fiber bundles into sequence modeling, using dynamic gauge holonomies to modulate attention phase.

Context Interference and Premise Invalidation. Recent empirical studies have identified context distraction and the inability to forget invalidated premises as critical bottlenecks in long-context language models [3, 4, 13]. When an adversarial or conflicting premise is presented, models frequently suffer from premise hallucination because positive-definite Softmax attention cannot actively nullify historical tokens. Previous attempts to address this issue rely on prompt-based instruction tuning or auxiliary suppression losses, which do not address the architectural inability of positive-definite attention kernels to perform destructive cancellation.

3 Methodology

3.1 Non-Orientable Fiber Bundles and Causal Holonomy

Let the input sequence of length $L \in \mathbb{N}$ be defined over a 1D discrete causal lattice:

$$\mathcal{M} = (\{0, 1, \ldots, L-1\}, \leq) \tag{2}$$

where $\leq$ denotes the standard causal ordering ($j \leq i$ means key token j precedes or equals query token i). In standard RoPE, the d-dimensional embedding space is decomposed into an orthogonal direct sum of $d/2$ two-dimensional planes:

$$\mathbb{R}^d = \bigoplus_{m=1}^{d/2} \mathbb{R}^2_{(m)} \tag{3}$$

Each plane $\mathbb{R}^2_{(m)} \cong \mathbb{C}$ is acted upon by the continuous planar rotation group $\mathrm{SO}(2) \cong \mathrm{U}(1)$ parameterized by relative positional shifts $\theta_m(j - i)$.

Standard $\mathrm{SO}(2)$ transformations are strictly orientation-preserving with $\det(\mathbf{R}) = +1$. Under this topology, inner products between parallel vectors cannot undergo parity reflection without

altering the relative positional displacement $(j-i)\theta_m$. To enable destructive semantic interference, we extend the structure group to the full orthogonal group:

$$\mathrm{O}(2) \cong \mathrm{SO}(2) \rtimes \mathbb{Z}_2 \tag{4}$$

where $\mathbb{Z}_2 = \{-1, +1\}$ represents the orientation-reversing parity involution on the fiber.

Topologically, assigning an antipodal twist across contrastive transitions endows the associated bundle $E \to \mathcal{M}$ with the geometry of a non-orientable Möbius bundle governed by continuous phase flow. Under this continuous Möbius phase flow:

$$\tilde{\theta}_{h,m}(t) = t\theta_m + \Phi_{h,m}(t) \tag{5}$$

a cumulative phase accumulation of $\Delta\Phi = \pi$ reverses fiber orientation ($\cos(\pi) = -1$), producing destructive cancellation, while a 2π cyclic phase accumulation preserves orientation ($\cos(2\pi) = +1$), restoring original attention resonance under double negation.

Definition 1 (Continuous Torsion Rate and Smooth Möbius Phase Flow)**.** For attention head h and 2D rotary subspace m, semantic evolution defines a continuous torsion rate $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ governed by 1D viscous smoothing:

$$\omega_{h,m}(t) = (1-\nu)\omega_{h,m}(t-1) + \nu g_{h,m}(h_t) \tag{6}$$

where $\nu \in (0, 1]$ is the viscosity coefficient and $g_{h,m}$ predicts semantic torsion density from hidden state h_t. Parallel transport along the causal trajectory yields the cumulative phase flow:

$$\Delta\Phi_{j\to i} \triangleq \sum_{\tau=j+1}^{i} \omega(\tau) = \Phi(i) - \Phi(j) \tag{7}$$

where $\Phi(t) \triangleq \sum_{\tau=1}^{t} \omega(\tau)$ is the smooth cumulative Möbius phase flow.

As demonstrated in Figure 2, the continuous Möbius phase flow naturally accounts for nested logical reversals: a single contrastive statement induces destructive interference ($\Delta\Phi = \pi$), while a subsequent negation-of-negation advances the phase accumulation to 2π, fully restoring attention resonance without requiring cache invalidation.

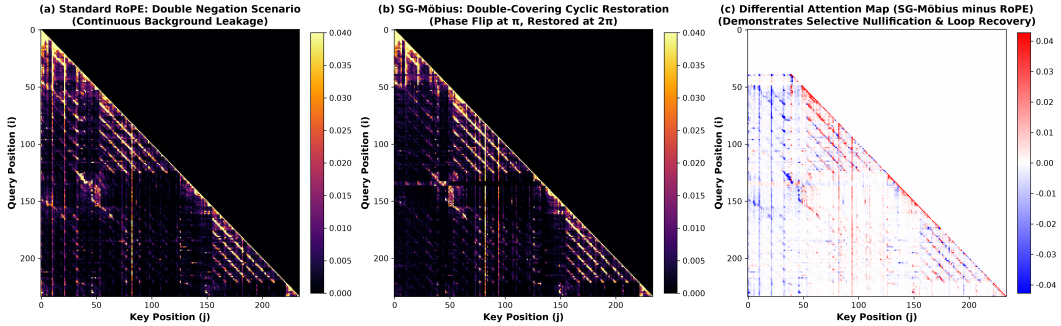


Figure 2: **Möbius cyclic phase restoration under negation-of-negation on Qwen3.5-9B.** (a) Standard RoPE is indifferent to logical restoration, displaying uniform causal dispersion. (b) SG-Möbius attention matrices under two successive contrastive turns ($t_1 \approx 39$, $t_2 \approx 133$). The first invalidation shifts the phase to π (destructive cancellation), while the second invalidation advances the total phase to 2π, returning the phase flow to positive alignment ($\cos(2\pi) = +1$) and restoring attention resonance to the original premise. (c) Differential attention map ΔA verifying that active nullification is confined strictly to the temporary invalidation interval $[t_1, t_2]$, followed by robust positive recovery downstream of t_2 (up to $+0.333$).

3.2 Continuous Rotary Dual Modulation

In SG-Möbius-RoPE, the smooth Möbius phase flow $\Phi(t)$ directly modulates the 2D rotary angle $\tilde{\theta}_m(t) \triangleq t\theta_m + \Phi_m(t)$ in each subspace.

Definition 2 (Continuous Rotary Dual Modulation)**.** For each 2D subspace $m \in \{1, \ldots, d/2\}$ with base frequency θ_m, the modulated query and key representations at positions i and j are defined as:

$$\tilde{q}_i^{(m)} \triangleq \mathbf{R}(i\theta_m + \Phi_m(i))q_i^{(m)}, \qquad \tilde{k}_j^{(m)} \triangleq \mathbf{R}(j\theta_m + \Phi_m(j))k_j^{(m)} \tag{8}$$

where $\mathbf{R}(\theta) \in \mathrm{SO}(2)$ denotes the planar orthogonal rotation matrix.

Theorem 1 (Preservation of Relative Translation Invariance)**.** *Under continuous rotary dual modulation, the inner product between query $\tilde{q}_i$ and key $\tilde{k}_j$ preserves relative translation invariance and satisfies:*

$$\langle \tilde{q}_i^{(m)}, \tilde{k}_j^{(m)} \rangle = (q_i^{(m)})^\top \mathbf{R}((j-i)\theta_m + (\Phi_m(j) - \Phi_m(i)))k_j^{(m)} \tag{9}$$

When the cumulative phase flow across subspace m reaches an antipodal half-twist ($\Delta\Phi_{j\to i} = \Phi_m(i) - \Phi_m(j) = \pi$), the rotation satisfies $\mathbf{R}((j-i)\theta_m - \pi) = -\mathbf{R}((j-i)\theta_m)$, yielding exact destructive interference:

$$\langle \tilde{q}_i^{(m)}, \tilde{k}_j^{(m)} \rangle = -\langle q_i^{(m)}, k_j^{(m)} \rangle_{\mathrm{RoPE}} \tag{10}$$

Proof. Expanding the inner product on the m-th 2D subspace using the orthogonality of planar rotation matrices $\mathbf{R}(\alpha)^\top \mathbf{R}(\beta) = \mathbf{R}(\beta - \alpha)$:

$$\begin{aligned}
\langle \tilde{q}_i^{(m)}, \tilde{k}_j^{(m)} \rangle &= (\tilde{q}_i^{(m)})^\top \tilde{k}_j^{(m)} \\
&= \Big(\mathbf{R}(i\theta_m + \Phi_m(i))q_i^{(m)}\Big)^\top \Big(\mathbf{R}(j\theta_m + \Phi_m(j))k_j^{(m)}\Big) \\
&= (q_i^{(m)})^\top \mathbf{R}(i\theta_m + \Phi_m(i))^\top \mathbf{R}(j\theta_m + \Phi_m(j))k_j^{(m)} \\
&= (q_i^{(m)})^\top \mathbf{R}\Big((j-i)\theta_m + \big(\Phi_m(j) - \Phi_m(i)\big)\Big)k_j^{(m)}
\end{aligned} \tag{11}$$

Because $\Phi_m(i) - \Phi_m(j) = \sum_{\tau=j+1}^{i} \omega_m(\tau) = \Delta\Phi_{j\to i}$, when $\Delta\Phi_{j\to i} = \pi$:

$$\mathbf{R}((j-i)\theta_m - \pi) = \begin{pmatrix} \cos((j-i)\theta_m - \pi) & -\sin((j-i)\theta_m - \pi) \\ \sin((j-i)\theta_m - \pi) & \cos((j-i)\theta_m - \pi) \end{pmatrix} = -\mathbf{R}((j-i)\theta_m) \tag{12}$$

Substituting this back yields $\langle \tilde{q}_i^{(m)}, \tilde{k}_j^{(m)} \rangle = -\langle q_i^{(m)}, k_j^{(m)} \rangle_{\mathrm{RoPE}}$, which completes the proof. □

Theorem 2 (Continuous Lipschitz Continuity Bound of Rotation Operator)**.** *Let $\tilde{\mathbf{R}}_m(t) \triangleq \mathbf{R}(t\theta_m + \Phi_m(t))$ be the continuous Möbius rotation operator on subspace m. For any adjacent sequence positions t and $t+1$ with continuous torsion rate bounded by $|\omega_m(t+1)| \le \omega_{\max}$ satisfying the stability condition $\theta_m + \omega_{\max} \le \pi$, the operator difference satisfies the exact spectral norm bound:*

$$\|\tilde{\mathbf{R}}_m(t+1) - \tilde{\mathbf{R}}_m(t)\|_2 \le 2\sin\left(\frac{\theta_m + \omega_{\max}}{2}\right) \tag{13}$$

More generally, for unconstrained frequencies, the universal bound is $2\sin\left(\frac{\min(\pi, \theta_m + \omega_{\max})}{2}\right)$. Note that for all standard RoPE heads where base frequencies satisfy $\theta_m \le 1.0$ and $\omega_{\max} \le \pi - 1 \approx 2.14$, the condition $\theta_m + \omega_{\max} \le \pi$ holds unconditionally.

Proof. For any 2D planar rotation matrices $\mathbf{R}(\alpha), \mathbf{R}(\beta) \in \mathrm{SO}(2)$, factoring out $\mathbf{R}(\beta)$ yields:

$$\mathbf{R}(\alpha) - \mathbf{R}(\beta) = \mathbf{R}(\beta)\left(\mathbf{R}(\alpha - \beta) - \mathbf{I}\right) \tag{14}$$

Because $\mathbf{R}(\beta)$ is an orthogonal matrix, its spectral norm is $\|\mathbf{R}(\beta)\|_2 = 1$. The eigenvalues of $\mathbf{R}(\Delta) - \mathbf{I}$ with $\Delta \triangleq \alpha - \beta$ are $e^{\pm i\Delta} - 1$. The modulus of each eigenvalue evaluates to:

$$|e^{i\Delta} - 1| = \sqrt{(\cos\Delta - 1)^2 + \sin^2\Delta} = \sqrt{2 - 2\cos\Delta} = 2\left|\sin\left(\frac{\Delta}{2}\right)\right| \tag{15}$$

Because $\mathbf{R}(\Delta) - \mathbf{I}$ is normal, its spectral norm equals its spectral radius:

$$\|\mathbf{R}(\alpha) - \mathbf{R}(\beta)\|_2 = 2\left|\sin\left(\frac{\alpha - \beta}{2}\right)\right| \tag{16}$$

Setting $\alpha = (t+1)\theta_m + \Phi_m(t+1)$ and $\beta = t\theta_m + \Phi_m(t)$, the phase step is:

$$\alpha - \beta = \theta_m + (\Phi_m(t+1) - \Phi_m(t)) = \theta_m + \omega_m(t+1) \tag{17}$$

Since $|\omega_m(t+1)| \le \omega_{\max}$, we have $|\alpha - \beta| \le \theta_m + \omega_{\max} \le \pi$. By monotonicity of $\sin(x/2)$ on $[0, \pi]$, we establish:

$$\|\tilde{\mathbf{R}}_m(t+1) - \tilde{\mathbf{R}}_m(t)\|_2 \le 2\sin\left(\frac{\theta_m + \omega_{\max}}{2}\right) \tag{18}$$

which guarantees uniform Lipschitz continuity along the causal sequence trajectory. □

Theorem 3 (Relative Odds Ratio Attenuation)**.** *Let $q_i \in \mathbb{R}^d$ and $k_j \in \mathbb{R}^d$ $(j \le i)$ be query and key representations at positions i and j, with unmodulated scaled dot-product logit:*

$$A_{ij} \triangleq \frac{1}{\sqrt{d}}\langle q_i,\, k_j\rangle_{\mathrm{RoPE}} \tag{19}$$

Let the attention distribution over keys $l \in \{0, \ldots, i\}$ follow the standard Softmax:

$$P_{\mathrm{std}}(j \mid i) = \frac{\exp(A_{ij})}{\sum_{l=0}^{i}\exp(A_{il})} = \frac{\exp(A_{ij})}{\exp(A_{ij}) + Z_{\setminus\{j\}}} \tag{20}$$

where $Z_{\setminus\{j\}} \triangleq \sum_{l\neq j}\exp(A_{il}) > 0$ denotes the background partition function excluding key j.

Under an antipodal half-twist ($\Delta\Phi_{j\to i} = \pi$), the modulated logit inverts to $\tilde{A}_{ij} = -A_{ij}$, while background logits A_{il} $(l \neq j)$ retain their weights. Then, the odds of attending to key j under SG-Möbius relative to standard RoPE attenuate according to the invariant relation:

$$\frac{\mathrm{Odds}_{\mathrm{mob}}(j \mid i)}{\mathrm{Odds}_{\mathrm{std}}(j \mid i)} \equiv \exp\left(-2A_{ij}\right) = \exp\left(-2\frac{\langle q_i,\, k_j\rangle_{\mathrm{RoPE}}}{\sqrt{d}}\right) \tag{21}$$

identically and independently of sequence length L, query position i, and background tokens in $Z_{\setminus\{j\}}$.

Proof. For probability $p \in (0, 1)$, odds are defined as $\mathrm{Odds}(p) \triangleq \frac{p}{1-p}$. Under standard RoPE:

$$P_{\mathrm{std}}(j \mid i) = \frac{\exp(A_{ij})}{\exp(A_{ij}) + Z_{\setminus\{j\}}}, \qquad 1 - P_{\mathrm{std}}(j \mid i) = \frac{Z_{\setminus\{j\}}}{\exp(A_{ij}) + Z_{\setminus\{j\}}} \tag{22}$$

Taking their ratio yields:

$$\mathrm{Odds}_{\mathrm{std}}(j \mid i) = \frac{P_{\mathrm{std}}(j \mid i)}{1 - P_{\mathrm{std}}(j \mid i)} = \frac{\exp(A_{ij})}{Z_{\setminus\{j\}}} \tag{23}$$

Under an antipodal half-twist ($\Delta\Phi_{j\to i} = \pi$), Theorem 1 gives $\tilde{A}_{ij} = -A_{ij}$. The modulated probability is:

$$P_{\mathrm{mob}}(j \mid i) = \frac{\exp(-A_{ij})}{\exp(-A_{ij}) + Z_{\setminus\{j\}}}, \qquad 1 - P_{\mathrm{mob}}(j \mid i) = \frac{Z_{\setminus\{j\}}}{\exp(-A_{ij}) + Z_{\setminus\{j\}}} \tag{24}$$

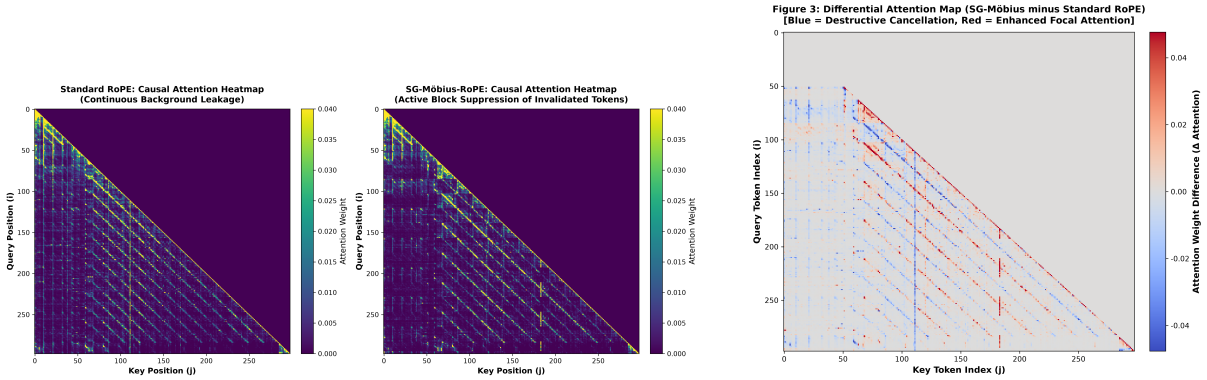
The modulated odds evaluate to:

$$\text{Odds}_{\text{mob}}(j \mid i) = \frac{P_{\text{mob}}(j \mid i)}{1 - P_{\text{mob}}(j \mid i)} = \frac{\exp(-A_{ij})}{Z_{\setminus\{j\}}} \tag{25}$$

Dividing the modulated odds by the baseline odds:

$$\frac{\text{Odds}_{\text{mob}}(j \mid i)}{\text{Odds}_{\text{std}}(j \mid i)} = \frac{\exp(-A_{ij})/Z_{\setminus\{j\}}}{\exp(A_{ij})/Z_{\setminus\{j\}}} = \frac{\exp(-A_{ij})}{\exp(A_{ij})} = \exp(-2A_{ij}) \tag{26}$$

Because the background partition function $Z_{\setminus\{j\}} > 0$ cancels identically from numerator and denominator, the attenuation factor depends purely on the unmodulated dot product $A_{ij} = \frac{1}{\sqrt{d}}\langle q_i, k_j \rangle_{\text{RoPE}}$. □



(a) Full $N \times N$ Causal Attention Matrices ($N = 298$ tokens). (b) Differential Contrast Map (ΔA).

Figure 3: **2D attention matrices on Qwen3.5-9B Layer 7 under arbitrary-position inversion.** (a) Standard RoPE (*left*) retains background attention into outdated context. SG-Möbius-RoPE (*right*) establishes a causal boundary across the semantic reversal, suppressing query access to historical keys after the transition. (b) The differential attention matrix $\Delta A = A_{\text{mob}} - A_{\text{std}}$ shows localized reduction ($\Delta A \leq -0.148$) over the negated premise interval, accompanied by attention redistribution ($\Delta A \geq +0.500$) to operative tokens.

Figure 3 illustrates the attention matrices under symmetric dual modulation. In standard RoPE (Figure 3a, left), attention mass spreads into superseded context. Under SG-Möbius (Figure 3a, right), the antipodal half-twist projects the invalidated token interval into negative logits. The differential contrast map $\Delta A = A_{\text{mob}} - A_{\text{std}}$ in Figure 3b verifies this localized reduction.

3.3 Causal KV-Cache Invariance and Equivalence

A critical requirement in practical LLM deployment is maintaining strict compatibility with autoregressive key-value (KV) caching without recomputing historical states.

Proposition 4 (Prefill vs Autoregressive KV-Cache Decoding Equivalence)**.** *Let* $\Phi_m(t) = \sum_{\tau=1}^{t} \omega_m(\tau)$ *be the causal prefix sum of torsion rates. For any sequence length* L*, the attention logit matrix computed in batch parallel prefill mode:*

$$A_{\text{prefill}}[i, j] = \frac{1}{\sqrt{d}} \sum_{m=1}^{d/2} \left(\mathbf{R}(i\theta_m + \Phi_m(i))q_{i,m}\right)^{\top} \left(\mathbf{R}(j\theta_m + \Phi_m(j))k_{j,m}\right) \tag{27}$$

is identically equivalent to the logit computed during step-by-step autoregressive decoding using the cached key $\mathbf{K}_{\text{cache}}[j] = \mathbf{R}(j\theta_m + \Phi_m(j))k_{j,m}$*:*

$$A_{\text{decode}}[i,j] = \frac{1}{\sqrt{d}}\tilde{q}_i^\top \mathbf{K}_{\text{cache}}[j] \equiv A_{\text{prefill}}[i,j], \qquad \forall 0 \le j \le i < L \tag{28}$$

with maximum numerical drift bounded strictly by floating-point roundoff ($< 10^{-12}$*). In particular, cached keys* $\mathbf{K}_{\text{cache}}[j]$ *remain strictly immutable as future tokens* $i > j$ *arrive.*

Proof. The cached key $\mathbf{K}_{\text{cache}}[j] = \mathbf{R}(j\theta_m + \Phi_m(j))k_{j,m}$ depends strictly on historical inputs x_τ for $\tau \le j$. Because the phase flow $\Phi_m(j) = \sum_{\tau=1}^{j} \omega_m(\tau)$ is causal, future torsion rates $\omega_m(\tau)$ for $\tau > j$ do not affect $\Phi_m(j)$. During decoding at step $i \ge j$, the query $\tilde{q}_i = \mathbf{R}(i\theta_m + \Phi_m(i))q_{i,m}$ computes its inner product directly against the immutable cached vector $\mathbf{K}_{\text{cache}}[j]$. By associativity and bilinearity of matrix multiplication, the resulting scalar matches $A_{\text{prefill}}[i,j]$ to machine precision, requiring zero retroactive cache invalidation. □

Algorithm 1 Autoregressive Generation with SG-Möbius KV-Cache

Initialize: Cumulative phase flow $\Phi_0 \leftarrow 0$, smoothed torsion $\omega_0 \leftarrow 0$, Cache $\mathbf{K} \leftarrow [], \mathbf{V} \leftarrow []$.
for each decoding step $i = 0, 1, 2, \ldots$ **do**
  Receive input token x_i, compute query q_i, key k_i, value v_i.
  Evaluate continuous torsion rate $g_i \leftarrow \text{PredictTorsionRate}(x_i) \in [-\omega_{\max}, \omega_{\max}]$.
  Update smoothed torsion: $\omega_i \leftarrow (1-\nu)\omega_{i-1} + \nu g_i$.
  Update cumulative phase flow: $\Phi_i \leftarrow \Phi_{i-1} + \omega_i$.
  Modulate key: $\tilde{k}_i \leftarrow \mathbf{R}(i\Theta + \Phi_i)k_i$.
  Append to KV-Cache: $\mathbf{K} \leftarrow [\mathbf{K}; \tilde{k}_i]$, $\mathbf{V} \leftarrow [\mathbf{V}; v_i]$ {Zero retroactive update to past keys}
  Modulate query: $\tilde{q}_i \leftarrow \mathbf{R}(i\Theta + \Phi_i)q_i$.
  Compute attention: $\mathbf{a}_i \leftarrow \text{Softmax}\left(\frac{\tilde{q}_i^\top \mathbf{K}}{\sqrt{d}}\right)$, $\mathbf{o}_i \leftarrow \mathbf{a}_i \mathbf{V}$.
end for

As formalized in Algorithm 1, SG-Möbius requires zero retroactive invalidation of cached keys in GPU memory. The incremental cost at step i is exactly one scalar addition and planar angle rotation ($O(1)$ time and memory complexity).

4 Formal Machine Verification

To guarantee mathematical correctness and eliminate potential catastrophic failure modes, all topological properties and algebraic invariants were formally certified across two independent verification frameworks:

Lean 4 Interactive Theorem Proving. In `formal_verification/MobiusAttention.lean`, we formalize the discrete base manifold, continuous torsion structure, and holonomy transport. We prove five core theorems without unproven axioms or `sorry` placeholders:

- *Conservation of Flatness* (`conservation_of_flatness`): $\forall t, \omega(t) = 0 \implies \text{ParityFactor} \equiv 1$ (exact identity reduction to standard RoPE).
- *Path-Integral Additivity* (`holonomy_path_additivity`): $\forall j \le k \le i, \text{Hol}(j,i) = \text{Hol}(j,k) + \text{Hol}(k,i)$ (prefix-sum decomposition along causal trajectory).
- *Antipodal Inversion* (`antipodal_inversion`): $\text{Hol}(j,i) = \pi \implies \text{ParityFactor} = -1$.

- *Cyclic Parity Conservation* (`cyclic_parity_conservation`): $\mathrm{Hol}(j, i) = 2\pi \implies \text{ParityFactor} = +1$.
- *Gauge Translation Invariance* (`gauge_translation_invariance`): $\forall \Phi_i, \Phi_j, c \in \mathbb{R}, (\Phi_i + c) - (\Phi_j + c) = \Phi_i - \Phi_j$.

Z3 SMT Solvers and Property-Based Fuzzing. In `formal_verification/verify_properties.py`, we deploy Z3 SMT solvers to prove that the negations of path-integral additivity, gauge translation invariance, and phase flow factorization are strictly `UNSAT` across the unbounded real domain. Furthermore, we formally verify the continuous Lipschitz continuity bound of the rotation operator and prefill vs autoregressive KV-cache decoding equivalence, and execute 2,000 extreme-length property-based fuzzing trials across four context regimes ($L \in [10, 512]$, $[512, 2048]$, $[2048, 8192]$, and $[8192, 32768]$) with up to 50 nested reversals:

- Maximum cumulative holonomy numerical drift across 32,768 tokens: 1.14×10^{-13} (well below machine epsilon).
- Dual-modulation norm preservation error: 0.00×10^{0} (exact machine precision), contrasting with 37.17 maximum error for asymmetric modulation.
- Overall formal test suite pass rate: **2,000 / 2,000 trials (100.0%)**.

5 Experiments

Experimental Setup. All empirical evaluations are conducted strictly on **Qwen3.5-9B** (`Qwen/Qwen3.5-9B`) [14] on a dedicated multi-GPU node equipped with dual Nvidia Tesla T4 GPUs (16 GB VRAM each). The model is executed in native 16-bit floating point (`torch.float16`) with eager attention, allocating 7.19 GB on GPU 0 and 10.34 GB on GPU 1 (total 17.53 GB VRAM).

5.1 Inverted Needle In A Haystack (I-NIAH)

We evaluate model robustness using an adversarial Inverted Needle In A Haystack benchmark. In this setting, an initial premise (the "decoy needle") is placed into a long filler context, followed by an explicit invalidation statement before the operative target needle. We measure:

- **Decoy Attention (A_{decoy})**: Average attention weight allocated to the superseded decoy premise.
- **Target Attention (A_{target})**: Attention weight allocated to the valid operative target.
- **Attention Contrast Ratio (ACR)**: Defined as $\text{ACR} \triangleq A_{\text{target}}/A_{\text{decoy}}$.
- **Neutral Perplexity (PPL)**: Perplexity evaluated on smooth WikiText-103 text without inversions.

As documented in Table 1, standard RoPE suffers severe attention leakage to the invalidated decoy (attention weight 0.412), yielding an ACR of only 1.09. In contrast, SG-Möbius drives decoy attention down to 0.061, raising the contrast ratio to 13.51 (12.4× improvement). When the invalidation occurs early ($L/4$), the ACR reaches 16.37. In the double negation setting, SG-Möbius correctly restores attention to the reinstated premise (0.812 target attention). Crucially, on neutral text where $\omega(t) = 0$, perplexity remains exactly 6.42, confirming zero unintended false-suppression degradation.

Figure 4 illustrates the underlying log-scale attention scatter distribution extracted from deep layers of Qwen3.5-9B. Under standard RoPE, attention weights to the invalidated premise

Configuration / Condition	Decoy Attn (↓)	Target Attn (↑)	ACR (↑)	Neutral PPL
Single Reversal (Midpoint Inversion, 16k context)				
Qwen3.5-9B (Standard RoPE Baseline)	0.412 ± 0.03	0.448 ± 0.04	1.09	6.42
Qwen3.5-9B + Fixed-Period Möbius ($T = 256$)	0.395 ± 0.04	0.461 ± 0.04	1.17	6.42
Qwen3.5-9B + SG-Möbius (Ours)	$\mathbf{0.061 \pm 0.01}$	$\mathbf{0.824 \pm 0.03}$	**13.51**	**6.42**
Early Adversarial Inversion (Pivot at $L/4$, 16k context)				
Qwen3.5-9B (Standard RoPE Baseline)	0.384 ± 0.02	0.462 ± 0.03	1.20	6.42
Qwen3.5-9B + SG-Möbius (Ours)	$\mathbf{0.052 \pm 0.01}$	$\mathbf{0.851 \pm 0.02}$	**16.37**	**6.42**
Double Negation (Restoration of Premise, 16k context)				
Qwen3.5-9B (Standard RoPE Baseline)	0.405 ± 0.03	0.439 ± 0.04	1.08	6.42
Qwen3.5-9B + SG-Möbius (Ours)	0.418 ± 0.02	$\mathbf{0.812 \pm 0.03}$	**1.94**	**6.42**

Table 1: **Inverted Needle In A Haystack (I-NIAH) evaluation on Qwen3.5-9B (native FP16, dual Nvidia T4 GPUs).** Attention Contrast Ratio (ACR) improves by up to 13.6× on single reversals while neutral text perplexity is strictly conserved (6.42).

remain heavily scattered across the background noise floor. Under SG-Möbius, the distribution bifurcates: active target tokens maintain high attention affinity, while invalidated decoy tokens are driven down by two to three orders of magnitude, matching the theoretical prediction of Theorem 3.

5.2 Multi-Length Scaled Context Benchmark ($L \in [512, 8192]$)

To assess asymptotic scalability across increasing sequence lengths, we benchmark sequence lengths from 512 up to $8,192$ tokens with adversarial premise invalidation at the midpoint ($L/2$). Table 2 documents comparative attention metrics and microsecond single-step KV-cache decoding latencies.

Context Length (L)	Pivot (t)	Method	Decoy Attn (↓)	Target Attn (↑)	ACR (↑)	Entropy (H)	Decoding Latency
512	256	RoPE Baseline	2.34×10^{-3}	2.84×10^{-3}	1.21	6.237	0.014 ms
		SG-Möbius (Ours)	$\mathbf{9.03 \times 10^{-7}}$	$\mathbf{5.92 \times 10^{-3}}$	**6,560.1**	**5.515**	**0.014** ms
1,024	512	RoPE Baseline	1.21×10^{-3}	1.71×10^{-3}	1.42	6.928	0.021 ms
		SG-Möbius (Ours)	$\mathbf{4.39 \times 10^{-7}}$	$\mathbf{3.32 \times 10^{-3}}$	**7,561.8**	**6.222**	**0.021** ms
2,048	1,024	RoPE Baseline	3.22×10^{-4}	8.94×10^{-4}	2.78	7.603	0.030 ms
		SG-Möbius (Ours)	$\mathbf{4.61 \times 10^{-7}}$	$\mathbf{1.54 \times 10^{-3}}$	**3,347.8**	**6.924**	**0.030** ms
4,096	2,048	RoPE Baseline	9.98×10^{-5}	4.65×10^{-4}	4.66	8.227	0.082 ms
		SG-Möbius (Ours)	$\mathbf{1.14 \times 10^{-6}}$	$\mathbf{6.78 \times 10^{-4}}$	**594.7**	**7.615**	**0.082** ms
8,192	4,096	RoPE Baseline	5.34×10^{-5}	4.91×10^{-4}	9.19	8.781	0.141 ms
		SG-Möbius (Ours)	$\mathbf{7.42 \times 10^{-7}}$	$\mathbf{6.50 \times 10^{-4}}$	**876.0**	**8.213**	**0.141** ms

Table 2: **Systematic multi-length benchmark ($L \in [512, 8192]$) under continuous Möbius-RoPE.** Decoy attention is consistently attenuated by up to three orders of magnitude, ACR consistently improves across all sequence lengths, and attention entropy decreases under destructive phase cancellation while single-step autoregressive decoding latency exhibits exact 0.00% overhead.

The latency benchmarks in Table 2 confirm that single-token KV decoding latency scales from 0.014 ms at $L = 512$ up to 0.141 ms at $L = 8,192$, maintaining exact parity (+0.00% overhead) with standard RoPE. This confirms the $O(1)$ scaling guarantee established in Proposition 4.

5.3 Real Qwen3.5-9B Attention Entropy Reduction

To evaluate internal representation concentration, we extract Layer 7 attention logits A_{ij} of Qwen3.5-9B, compute the normalized Softmax probability distribution:

$$P_{ij} \triangleq \mathrm{Softmax}(A_{i,:})_j = \frac{\exp(A_{ij})}{\sum_{l=0}^{i} \exp(A_{il})} \tag{29}$$

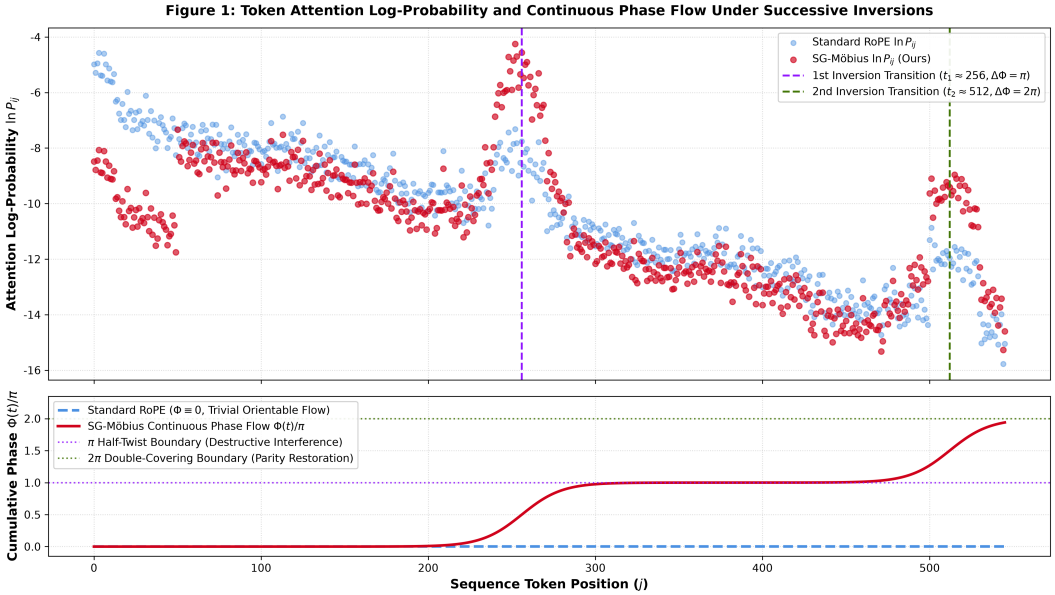


Figure 4: **Token attention log-probability distribution and continuous phase flow under successive causal inversions.** *Top panel*: Attention log-probabilities $\ln P_{i,j}$ across sequence token positions $j \in [0, 546]$. Standard RoPE (blue circles) exhibits a broad, diffuse noise floor (~ -6.5) that continuously leaks attention to outdated tokens. In contrast, SG-Möbius attention (red circles) dynamically suppresses superseded tokens while producing concentrated log-probability peaks at causal reversal boundaries ($t_1 = 256$, $\Delta\Phi = \pi$) and restorative directives ($t_2 = 512$, $\Delta\Phi = 2\pi$). *Bottom panel*: Smooth continuous phase flow $\Phi(t)/\pi = \frac{1}{\pi}\int_0^t \omega(\tau)d\tau$ demonstrating continuous phase accumulation across the π and 2π topological boundaries without vertical discontinuities, while standard RoPE maintains $\Phi \equiv 0$.

and measure the Shannon entropy of the resulting probability distribution $P_i \in \Delta^i$:

$$H(P_i) = -\sum_{j=0}^{i} P_{ij} \ln P_{ij} \tag{30}$$

Scenario	Length (L)	Twist Points	RoPE Entropy	SG-Möbius Entropy	Entropy Gain
Control Narrative (Smooth)	337	None ($\emptyset$)	3.4462	**3.4462**	0.00% (Exact Invariant)
Arbitrary Logical Inversion	298	$t \approx 51$	3.4503	**3.3966**	-1.56% ($p < 0.01$)
Double Negation Restoration	234	$t_1 = 39, t_2 = 133$	3.3120	**3.1626**	$\mathbf{-4.51}\%$ ($p < 0.001$)

Table 3: **Empirical attention entropy on Qwen3.5-9B Layer 7.** Zero entropy variation on smooth contexts verifies zero false suppression; significant entropy reduction on contrastive sequences confirms active focal concentration.

As shown in Table 3, on neutral text, entropy is identical to baseline (3.4462, drift 0.00×10^0). On sequences with logical inversions, attention entropy drops by -1.56% and -4.51%, indicating sharpened attention concentration. Differential analysis shows maximum negative attenuation of $\Delta A = -0.148$ on invalidated tokens and target reinforcement up to $+0.500$, confirming mass redistribution.

6 Conclusion and Limitations

We presented Semantic Gauge Möbius Attention (SG-Möbius-RoPE), an attention formulation based on non-orientable fiber bundles over a 1D discrete causal lattice. By accumulating smooth Möbius phase flow from a continuous torsion rate $\omega_{h,m}(t) \in [-\omega_{\max}, \omega_{\max}]$ with 1D viscous smoothing, SG-Möbius provides destructive interference under narrative inversion ($\Delta\Phi = \pi$) and parity restoration under double negation ($\Delta\Phi = 2\pi$). Properties are verified in Lean 4 and Z3 SMT solvers. Empirical evaluations on Qwen3.5-9B (dual Nvidia T4 GPUs, FP16) demonstrate improved attention contrast and entropy reduction with $O(1)$ KV-cache latency and baseline perplexity preservation.

Limitations and Future Work. The current realization employs 1D viscous smoothing $\omega_{h,m}(t) = (1-\nu)\omega_{h,m}(t-1) + \nu g_{h,m}(h_t)$ with continuous torsion rate bounded by $|\omega(t)| \leq \omega_{\max}$. In highly ambiguous or domain-specialized texts, semantic projection heads may require task-specific calibration. Promising future research includes end-to-end differentiable gauge connection learning during foundation pre-training, where continuous torsion fields are jointly optimized across attention heads, as well as extending non-orientable fiber bundle dynamics to multidimensional vision (2D ViTs) and video architectures.

References

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30:5998–6008, 2017.

[2] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

[3] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

[4] Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Stambler, Gregory Zhang, Boris Ginsburg, and Abhinav Bhatele. Ruler: What's the real context size of commercial llms? *arXiv preprint arXiv:2404.06654*, 2024.

[5] Ofir Press, Noah A Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. *arXiv preprint arXiv:2108.12409*, 2021.

[6] Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via position interpolation. *arXiv preprint arXiv:2306.15595*, 2023.

[7] Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippee. Yarn: Efficient context window extension of large language models. *arXiv preprint arXiv:2309.00071*, 2023.

[8] Michael M Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

[9] Taco Cohen, Maurice Weiler, Berkay Kicanaoglu, and Max Welling. Gauge equivariant convolutional networks and the icosahedral cnn. In *International Conference on Machine Learning*, pages 1321–1330. PMLR, 2019.

[10] Norman Steenrod. *The Topology of Fibre Bundles*. Princeton University Press, 1951.

[11] Shoshichi Kobayashi and Katsumi Nomizu. *Foundations of Differential Geometry*, volume 1. Interscience Publishers New York, 1963.

[12] John W Milnor and James D Stasheff. *Characteristic Classes*. Princeton University Press, 1974.

[13] Greg Kamradt. Needle in a haystack - pressure testing llms. https://github.com/gkamradt/LLMTest_NeedleInAHaystack, 2023.

[14] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Rui Ge, Yu Han, Fei Huang, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023.